

EDUCATION

University of North Carolina at Chapel Hill – Data Science and Statistics.

Aug 2023 -May 2027

RESEARCH INTERESTS

Statistical machine learning, predictive modelling, LLM reasoning and robustness, AI for decision systems, and data-driven models to reduce barriers in insurance, public benefits, and service delivery systems.

PROJECT EXPERIENCE

Dollars in Orbit — Carolina Data Challenge 2025 (2nd Place Winner)

September 2025

- Analysed BEA space economy data (2012–2023) to quantify how private and federal sectors contributed to the U.S. space output.
- Developed Excel dashboards and KPI visualisations showing GDP contribution and funding impact ($r \approx 0.59$).

Tweet Sentiments During the COVID-19 Lockdown

January 2025 – April 2025

- Built and evaluated Random Forest and SVM models to analyse relationships between Twitter sentiment and COVID-19 indicators.
- Preprocessed text and structured datasets for predictive modelling, achieving 92% R^2 for death prediction and 37% accuracy for sentiment classification.

Lovify

August 2024 – November 2024

- Built a web application using Spotify's API to analyse user listening behavior and generate insights from engineered audio features.
- Developed backend logic and feature-based classifications to model tempo, bass, genre, and sentiment patterns.

PUBLICATIONS

Smart City Surveillance System

June 2022

International Journal for Innovative Research in Multidisciplinary Field

- Developed a conceptual framework for integrating public-service data across hospitals, police, traffic, schools, and municipal systems to improve response coordination and public safety.
- Explored how system-level data integration could strengthen service delivery and urban decision-making.

IoT-based Smart Waste Collector

August 2022

International Journal of Innovative Research in Science, Engineering and Technology

- Designed an IoT-enabled waste monitoring concept using sensors, web-based reporting, and automated alerts to improve waste collection efficiency.
- Focused on using embedded systems and data visibility to support cleaner and more responsive urban infrastructure.

EXPERIENCE

PQLS – Tamil Nadu, India

January 2026 - Present

Business Analytics Intern

- Analysed 2,300 customer records across 35 variables using SQL and Excel to identify service patterns and improvement opportunities.
- Built Tableau dashboards to support reporting and data-driven business decisions.

Adroitts – Reston, VA

June 2025 - August 2025

Data Science Intern

- Designed Tableau dashboards to translate large marketing datasets into actionable insights, highlighting client KPIs for executive decision-making.

Carolina Housing – Chapel Hill, NC

August 2024 - Present

Resident Advisor

- Supported a residential community through conflict mediation, crisis response, and policy-based decision-making.

SKILLS

Python (Pandas, NumPy, scikit-learn) · R · SQL · Power BI · Tableau · Machine Learning (Predictive Modelling, Random Forest, SVM, Regression) · Web Development (Flask, React.js, APIs) · MS Excel